Modeling Monotonic and Cyclic Behavior of Granular Materials by Exponential Constitutive Function

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Abstract: It is still an open problem to develop a model with a uniform theoretical treatment under various loading conditions. This paper aims to formulate such a model for describing both monotonic and cyclic behaviors of granular materials. An exponential function was adopted to reproduce the stress-strain relation. Within the framework of endochronic models, the shear strain component was enriched with an absolute term to account for a reverse loading effect during shearing. The capacity of the basic model with four parameters for reproducing the basic features of granular materials was examined. Three numerical schemes for simulating undrained triaxial tests, drained triaxial tests under constant p' or under constant confining stress, and under both monotonic and cyclic loadings were established. Then, three modifications were carried out to enhance the model by introducing a nonlinear elasticity, a nonlinear stress dilatancy, and the critical state concept with four additional parameters. The enhanced version of the model showed good performances in simulating triaxial tests on Toyoura sand under various loading conditions: drained, undrained, constant p', constant confining stress, and monotonic and cyclic loadings. **DOI: 10.1061/(ASCE)EM.1943-7889.0001437.** *© 2018 American Society of Civil Engineers.*

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Introduction

In civil engineering, granular materials can serve as construction fills or foundation soils. Because of their complex behavior, numerous constitutive models have been developed. These models can be classified as (1) nonlinear hypoelastic models (e.g., Duncan and Chang 1970), (2) incrementally nonlinear models (e.g., Darve and Labanieh 1982; Darve 1990), (3) hypoplastic models (e.g., Wu et al. 1996; Niemunis and Herle 1997; Mašín and Khalili 2012), (4) elastoplastic models (e.g., Vermeer 1978; Jefferies 1993; Yu 1998; Gajo and Wood 1999; Taiebat and Dafalias 2008; Yao et al. 2004, 2008), and (5) micromechanics-based models (e.g., Chang and Hicher 2005; Yin et al. 2010, 2014; Chang et al. 2010; Nicot and Darve 2011; Wan and Pouragha 2015). In general, the first four categories are phenomenological models, which are commonly adopted in engineering practice due to their efficiency in finite-element analyses.

According to the mechanical behavior of granular materials, a phenomenological model should be able to describe the frictional behavior (asymptotic relationship between the stress ratio and the shear strain), the contractive or dilative behavior (shear-induced volume change), the critical state behavior (the unique ultimate state of a given material attainable in the $p'-q$ and the $e-p'$ planes for any initial state). The phenomenological models that can successfully capture all these mechanical features require usually more than 10 parameters with more or less complicated formulations. Therefore, it is still an open problem to develop a model with simple formulations and a small number of parameters.

This paper attempts to formulate a simple model with few parameters able to model both monotonic and cyclic behaviors of granular materials. The key formulation followed the framework of endochronic models adopting exponential function in which the absolute shear strain was used to account for a reverse loading effect in shearing. A stress-dilatancy relation was developed for the shear-induced volume change. The basic model was then examined on reproducing the basic features of granular materials by establishing three triaxial test simulation schemes. After that, a nonlinear elasticity, a nonlinear stress dilatancy, and the critical state concept were introduced to enhance the model. Finally, the predictive ability of the model was examined by simulating various triaxial tests on Toyoura sand under different loading conditions.

Model Based on an Exponential Function

Basic Function

Granular materials exhibit a nonlinear frictional behavior with an asymptotic relationship between the stress ratio and the shear strain, which is a key point in phenomenological constitutive modeling (Duncan and Chang 1970; Vermeer 1978; Jefferies 1993; Biarez and Hicher 1994). This frictional asymptotic response can be described by various mathematical functions, for example, the hyperbolic or exponential functions [see Figs. 1(a and b) for given values of a and b]

$$\eta = \frac{b\gamma}{1/a+\gamma} \quad \text{or} \quad \eta = b(1-e^{-a\gamma}) \tag{1}$$

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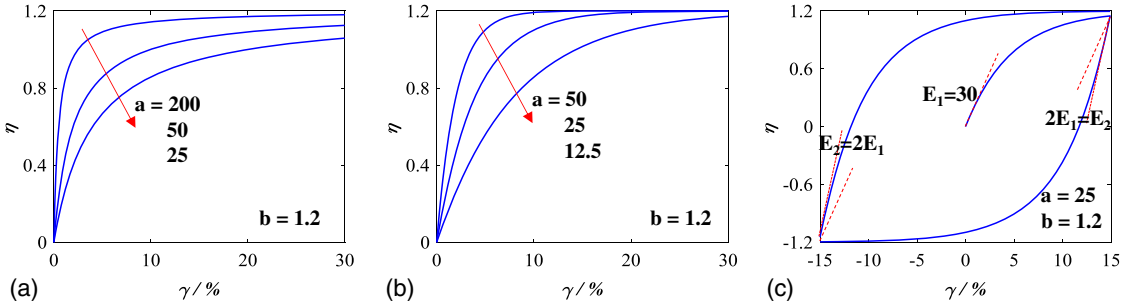

Fig. 1. Stress ratio versus shear strain described by (a) a hyperbolic function; (b) an exponential function; (c) a modified incremental form of the exponential function

In nonlinear constitutive modeling, the incremental form has to be considered and the aforementioned functions can be rewritten in an incremental form

$$\dot{\eta} = \frac{ab\dot{\gamma}}{(1+a\gamma)^2} \quad \text{or} \quad \dot{\eta} = ab\dot{\gamma}\left(1-\frac{\eta}{b}\right) \tag{2}$$

However, Eq. (1) or (2) is not able to describe the reversed asymptotic relationship between the stress ratio and the shear strain during unloading. To model the cyclic behavior of granular materials, the aforementioned equations should be extended to unloading and reloading conditions.

As suggested by Sandler (1978) for an endochronic model, the incremental form of the exponential function can be enriched by taking into account the absolute value of the shear strain increment

$$\dot{\eta} = ab\left(\dot{\gamma} - \frac{\eta}{b}|\dot{\gamma}|\right) \tag{3}$$

Then for the unloading, the shear strain increment changes its sign from positive to negative, which mathematically results in an unloading asymptotic curve with a twice initial stiffness of loading stage [see Fig. 1(c) for given values of a and b]. Thus, this formulation extends the function from loading to reverse loading, and thus makes this function applicable to describe the cyclic behavior. Therefore, Eq. (3) has been adopted as a basis for the development of the model, as presented in the following sections.

Basic Model under Triaxial Condition

Under triaxial condition, the stress ratio $\eta = q/p'$ is defined by the deviatoric stress q ($q = \sigma_a - \sigma_r$) and the mean effective stress p' [$p' = (\sigma_a' + 2\sigma_r')/3$], where σ_a and σ_r are the axial and radial stresses, respectively. Eq. (3) can be rewritten in the following form:

$$\dot{\eta} = G_\eta\left[\dot{\varepsilon}_d - \left(\frac{\eta}{M_p}\right)|\dot{\varepsilon}_d|\right] \tag{4}$$

where $\dot{\varepsilon}_d = 2(\dot{\varepsilon}_a - \dot{\varepsilon}_r)/3$ = deviatoric strain increment function of the axial and radial strain increments; M_p = strength ratio; and G_η = initial stiffness.

The stress ratio increment can be derived as

$$\eta = \frac{q}{p'} \Rightarrow \dot{\eta} = \frac{\dot{q}}{p'} - \frac{q}{p'^2}\dot{p}' = (\dot{q} - \eta\dot{p}')\frac{1}{p'} \tag{5}$$

Combining Eqs. (5) and (4), the incremental deviatoric stress is obtained as

$$\dot{q} = G_\eta\left[p'\dot{\varepsilon}_d - \left(\frac{q}{M_p}\right)|\dot{\varepsilon}_d|\right] + \frac{q}{p'}\dot{p}' \tag{6}$$

According to this equation, the initial slope of the curve $q - \varepsilon_d$, termed as three times the shear modulus G, can be obtained by assuming that q is very small for a small value of ε_d

$$3G = \frac{\partial q}{\partial \varepsilon_d} = G_\eta\left(p' - \frac{q}{M_p}\right) \doteq G_\eta p' \Rightarrow G_\eta = \frac{3G}{p'} \tag{7}$$

Substituting Eq. (7) into Eq. (6), the incremental deviatoric stress is updated as

$$\dot{q} = 3G\left[\dot{\varepsilon}_d - \left(\frac{q}{p'M_p}\right)|\dot{\varepsilon}_d|\right] + \frac{q}{p'}\dot{p}' \tag{8}$$

A volumetric strain is usually induced during shearing, known as stress-dilatancy behavior. The stress-dilatancy relationship suggested by Roscoe et al. (1963) and verified by Yin and Chang (2013) has been modified to satisfy both loading and unloading conditions

$$\frac{\dot{\varepsilon}_v^{in}}{\dot{\varepsilon}_d} = M_{pt} - \frac{q}{p'} \Rightarrow \frac{\dot{\varepsilon}_v^{in}}{\dot{\varepsilon}_d} = \frac{\dot{\varepsilon}_d}{|\dot{\varepsilon}_d|}M_{pt} - \frac{q}{p'} \tag{9}$$

where $\dot{\varepsilon}_v^{in}$ = shear strain-induced volumetric strain increment. In Eq. (9) $\dot{\varepsilon}_d > 0$ during loading results in $\dot{\varepsilon}_d/|\dot{\varepsilon}_d| = 1$ and a linear relationship between $\dot{\varepsilon}_v^{in}/\dot{\varepsilon}_d$ and q/p' with an intersection value of M_{pt} at $q/p' = 0$ [Fig. 2(a)]; and $\dot{\varepsilon}_d < 0$ during unloading results in $\dot{\varepsilon}_d/|\dot{\varepsilon}_d| = -1$ and a linear relationship between $\dot{\varepsilon}_v^{in}/\dot{\varepsilon}_d$ and q/p' with an intersection value of $-M_{pt}$ at $q/p' = 0$ [Fig. 2(a)], which captured the experimental phenomenon of stress dilatancy during cyclic loadings (Pradhan 1990). For a neutral loading with $\dot{\varepsilon}_d = 0$, $\dot{\varepsilon}_d/|\dot{\varepsilon}_d| = 1$ can be imposed to keep the original form of the equation. The proposed model is under the framework of endochronic models using directly the sign of shear strain increment to judge loading or unloading, which is different from general plasticity. The phase-transformation stress ratio M_{pt} has been defined in relation to the dilatancy angle of the granular material. Eq. (9) could be simplified as

$$\dot{\varepsilon}_v^{in} = \frac{(\dot{\varepsilon}_d)^2}{|\dot{\varepsilon}_d|}M_{pt} - \frac{q}{p'}\dot{\varepsilon}_d = M_{pt}|\dot{\varepsilon}_d| - \frac{q}{p'}\dot{\varepsilon}_d \tag{10}$$

The mean effective stress increment can be computed as

$$\dot{p}' = K(\dot{\varepsilon}_v - \dot{\varepsilon}_v^{in}) = K\left(\dot{\varepsilon}_v - M_{pt}|\dot{\varepsilon}_d| + \frac{q}{p'}\dot{\varepsilon}_d\right) \tag{11}$$

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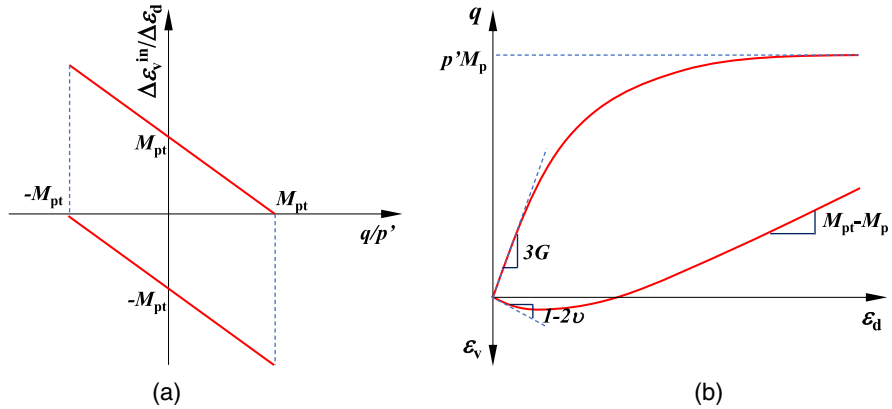

Fig. 2. Schematic plot of a typical drained triaxial test with the definition of the model parameters

where K = bulk modulus; and $\dot{\varepsilon}_v$ = volumetric strain increment. Under axisymmetric condition, $\dot{\varepsilon}_v$ is function of the axial and radial strain increments: $\dot{\varepsilon}_v = \dot{\varepsilon}_a + 2\dot{\varepsilon}_r$.

Finally, the basic model is completely defined by Eqs. (8) and (11). The constitutive equations are nonsymmetric, and the degree of nonsymmetry sharply changes upon strain reversal.

Summary of the Basic Model Parameters

The basic model includes four parameters directly related to those present in the Mohr–Coulomb model [Fig. 2(b)], summarized as follows:

- The bulk modulus K and the shear modulus G, which are equivalent to the use of the Young's modulus E and Poisson's ratio υ [$E = 9KG/(3K+G)$, $\upsilon = (3K-2G)/(6K+2G)$] as in the Mohr–Coulomb model, or the use of G and υ measured from drained triaxial tests. Because the small strain stiffness was not considered in this model, the elastic parameters can be calibrated directly from conventional tests instead of specified small strain tests.
- The strength ratio M_p depends on the friction angle of the material, as in the Mohr–Coulomb model: $M_p = 6\sin\phi_c/(3-\sin\phi_c)$.
- The phase-transformation stress ratio M_{pt} depends on the dilation angle ψ as in the Mohr–Coulomb model: $M_{pt} = M_p - 2\sin\psi/(1-\sin\psi)$.

A cohesion can also be easily introduced in the model if p' in Eqs. (8) and (11) is replaced by $p' + 2c/M_p$, with c the cohesion as in the Mohr–Coulomb model.

Simulation Schemes and Predictive Performance of the Basic Model

In this section, three simulation schemes for undrained triaxial tests and drained triaxial tests under constant p' and under constant confining stress, for both monotonic and cyclic loadings, have been established. Then, for a given set of parameters, the predictive performance of the basic model was examined according to the results of experimental investigations.

Undrained Triaxial Test Simulations

Eqs. (8) and (11) can be used directly for simulating the tests under full strain control. For example, $\dot{\varepsilon}_d = 0$ is imposed for isotropic compression implied by Eqs. (8) and (11); $\dot{\varepsilon}_v = 0$ is imposed for undrained shearing. The undrained triaxial test was simulated by a given set of parameters, as shown in Figs. 3(a and b). Different values of the phase-transformation stress ratio led to different typical behaviors of the granular material, depending on its initial density.

To simulate the undrained cyclic triaxial test under stress control with given values of the maximum and minimum deviatoric stresses, a deviatoric strain increment was first imposed with null volumetric strain increment. The deviatoric strain reversal was then conducted when the current deviatoric stress exceeded the maximum or minimum deviatoric stress. In other words, this stress control test was numerically conducted under strain control. With this technique, an undrained cyclic triaxial test was simulated by a given set of parameters, as shown in Figs. 3(c and d). Again, different values of the phase-transformation stress ratio led to different typical behaviors of the granular material (i.e., cyclic mobility for dense sand and cyclic liquefaction for loose sand).

Drained Constant-p' Test Simulations

Under the drained constant-p' test condition, the incremental mean effective stress is null. Thus, the volumetric strain and the deviatoric stress can be obtained from Eqs. (8) and (11)

$$\dot{p}' = 0 \Rightarrow \left\{ \begin{array}{l} \dot{\varepsilon}_v = M_{pt}|\dot{\varepsilon}_d| - \frac{q}{p'}\dot{\varepsilon}_d \\ \dot{q} = 3G\left[\dot{\varepsilon}_d - \left(\frac{q}{p'M_p}\right)|\dot{\varepsilon}_d|\right] \end{array} \right\} \tag{12}$$

A drained constant-p' test can, therefore, be simulated by imposing the incremental deviatoric strain in Eq. (12).

Tests under monotonic loading were first simulated by a given set of parameters, as shown in Figs. 4(a and b). Different values of the phase-transformation stress ratio gave different typical contractive and dilative behaviors of the granular material, corresponding to dense or loose sand. However, the calculated deviatoric stress appeared independent of the stress dilatancy, as it is the case for many simple models, such as the Mohr–Coulomb model, the Duncan–Chang model (Duncan and Chang 1970), the basic hypoplastic model of Wu and Kolymbas (1996), the hardening soil model of Schanz et al. (1999). This aspect needs, therefore, to be improved.

The simulation technique for a cyclic test under stress control with given values of the maximum and minimum deviatoric stresses is similar to the one used for an undrained test, as presented in the previous section. Another cyclic test under strain control condition was simulated by imposing directly the strain increment between the maximum and minimum imposed strain values. The deviatoric strain control cyclic test was simulated with a given set of parameters, as shown in Figs. 4(c and d). Again, different values of the phase-transformation stress ratio gave different typical behaviors of the granular material, that is, a progressive dilation

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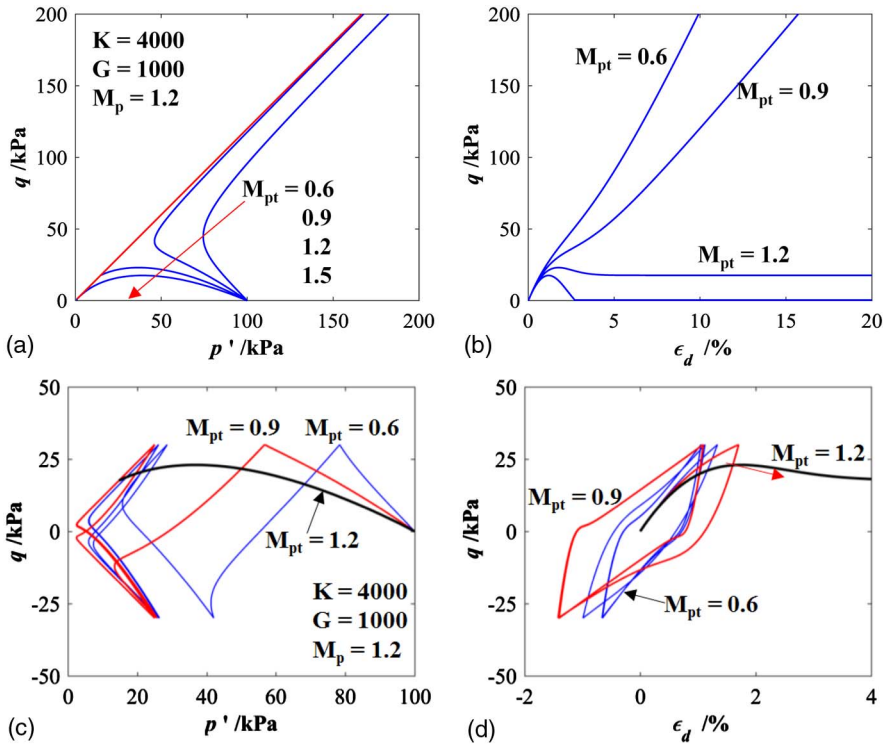

Fig. 3. Simulated results of undrained triaxial tests: (a) stress path under monotonic loading; (b) deviatoric stress strain under monotonic loading; (c) stress path under cyclic loading; (d) deviatoric stress strain under cyclic loading

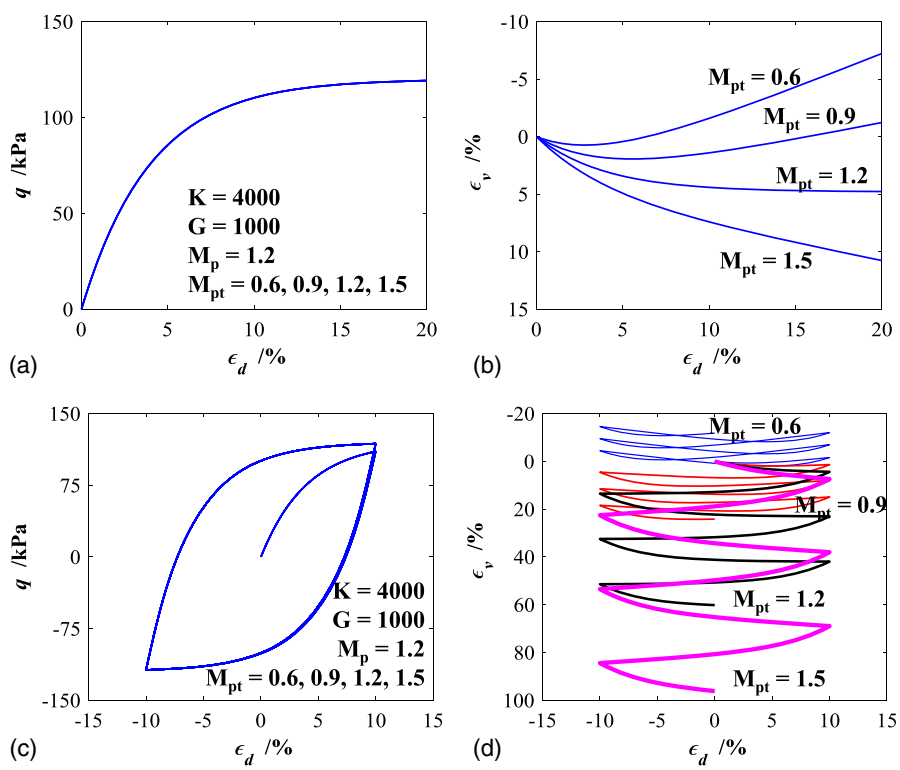

Fig. 4. Simulated results of drained constant-p' tests: (a) deviatoric stress strain under monotonic loading; (b) volumetric strain versus deviatoric strain under monotonic loading; (c) deviatoric stress strain under cyclic loading; (d) volumetric strain versus deviatoric strain under cyclic loading

for a dense sand and a progressive contraction for a loose sand. Similar to the results obtained for a monotonic test, the deviatoric stress-strain response appeared independent of the stress dilatancy, which needs also to be improved.

Conventional Drained Triaxial Test Simulations

In a conventional drained triaxial test, the confining stress is maintained constant. Thus, the ratio between the incremental deviatoric stress and the incremental mean effective stress is equal to 3.

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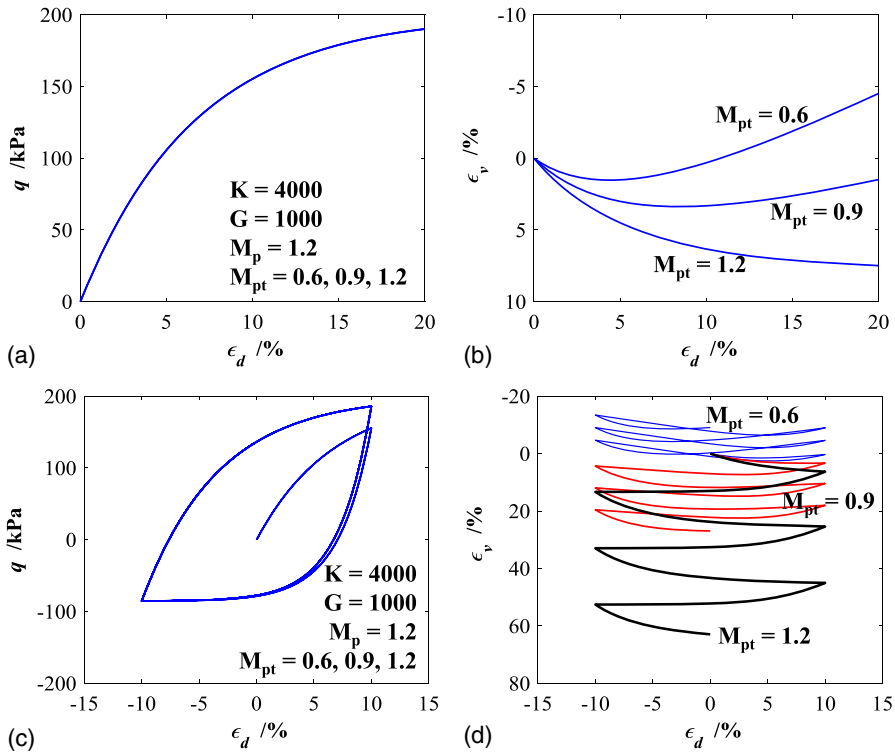

Fig. 5. Simulated results of conventional drained triaxial tests: (a) deviatoric stress strain under monotonic loading; (b) volumetric strain versus deviatoric strain under monotonic loading; (c) deviatoric stress strain under cyclic loading; (d) volumetric strain versus deviatoric strain under cyclic loading

From Eqs. (8) and (11), the volumetric strain and the deviatoric stress can be obtained as follows:

$$\dot{q} = 3G\left[\dot{\varepsilon}_d - \left(\frac{q}{p'M_p}\right)|\dot{\varepsilon}_d|\right] + \frac{q}{p'}\dot{p}' \overset{\dot{q}=3\dot{p}'}{\longrightarrow} \dot{q} = 3G\left[\dot{\varepsilon}_d - \left(\frac{q}{p'M_p}\right)|\dot{\varepsilon}_d|\right] \Big/ \left(1 - \frac{q}{3p'}\right) \quad (13)$$

$$\dot{p}' = \frac{\dot{q}}{3} \quad (14)$$

$$\dot{\varepsilon}_v = \frac{\dot{p}'}{K} + M_{pt}|\dot{\varepsilon}_d| - \frac{q}{p'}\dot{\varepsilon}_d \quad (15)$$

Then, the conventional drained triaxial test can be simulated by imposing an incremental deviatoric strain, following the order of Eqs. (13)–(15).

As for the constant-p' tests, monotonic and cyclic loadings were simulated, based on a given set of parameters, as shown in Fig. 5. The same conclusions as for the constant-p' test could be made, except for the cyclic test, which presented a less symmetric behavior between the unloading and reloading phases.

Enhancement of the Model

Nonlinear Elasticity

To introduce the dependency of the elastic stiffness on the density and the mean effective stress, the expression of the hypoelastic shear modulus G suggested by Richart et al. (1970) has been adopted

$$G = G_0 \frac{(2.97 - e)^2}{(1 + e)}\left(\frac{p'}{p_{at}}\right)^n,$$
$$K = K_0 \frac{(2.97 - e)^2}{(1 + e)}\left(\frac{p'}{p_{at}}\right)^n \quad (16)$$

with two parameters K_0 and G_0 instead of K and G, and one additional parameter n (typically 0.5–0.7). The atmospheric pressure $p_{at} = 101.325$ kPa was chosen as the reference stress.

To identify the change due to the consideration of a nonlinear elasticity, monotonic constant-p' tests were simulated with a given set of parameters and three different initial void ratios, as shown in Fig. 6. The results demonstrated that an increase in the void ratio led to a smaller initial slope of the stress-strain curve and a less dilative behavior, in agreement with the general experimental phenomena.

Critical State Concept

Several formulas for expressing the critical state concept have been proposed. The most typical one is the linear formula in the $e - \log p'$ plane. The relationship has traditionally been written as follows:

$$e_c = e_{\text{ref}} - \lambda \ln\left(\frac{p'}{p_{\text{ref}}}\right) \quad (17)$$

where e_{ref} = reference void ratio corresponding to a reference mean effective stress p_{ref}; λ = slope of the critical state line (CSL) in the $e - \log p'$ plane. Therefore, two parameters (e_{ref} and λ) are required for the definition of the CSL. The advantage of this formula is the simplicity of its form. However, experimental results have shown that the CSL is not always linear in the $e - \log p'$ plane,

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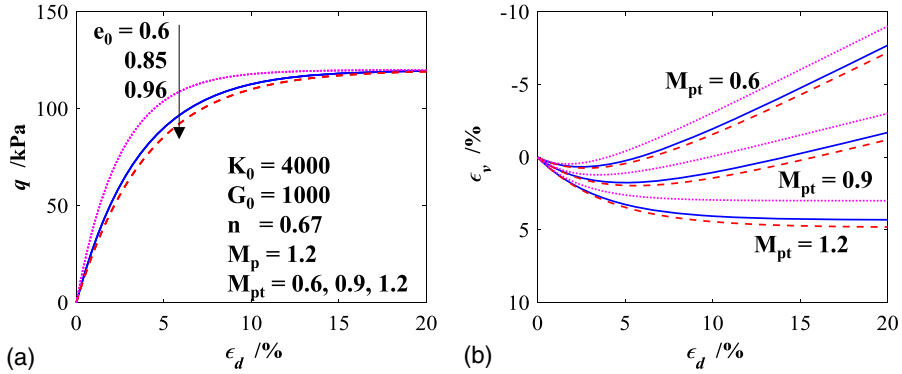

Fig. 6. Simulated results of drained constant-p' tests with two different initial void ratios: (a) deviatoric stress strain; (b) volumetric strain versus deviatoric strain

and, mathematically, the critical void ratio e_c could become negative for high stress levels, which is physically incorrect. The very high stress level still exists in geotechnical structures, like pile tip during its installation.

More recently, another formula has been proposed by Li and Wang (1998) who assumed a nonlinear CSL in the $e - \log p'$ plane, representing an extension of the linear formula with one additional parameter ξ

$$e_c = e_{c0} - \lambda\left(\frac{p'}{p_{at}}\right)^{\xi} \tag{18}$$

where the additional parameter ξ controls the nonlinearity of the CSL, giving a more flexible and accurate description according to experimental data, especially for very low to moderate stress levels. However, for high stress levels the positiveness of critical void ratio e_c could not be guaranteed, and possibly cause numerical problem of some local elements in finite-element modeling. To overcome this difficulty, Gudehus (1997) suggested a third formula of the CSL, which is also a nonlinear formula but with an s form by considering an ultimate critical void ratio at very high stress levels. It can be expressed as follows:

$$e_c = e_{cu} + (e_{c0} - e_{cu})\exp\left[-\left(\frac{p'}{p_{at}\cdot\lambda}\right)^{\xi}\right] \tag{19}$$

where e_{cu} = critical void ratio when $p' \to \infty$. This expression eliminates the possibility of a negative value of the critical void ratio at high stress levels. However, two more parameters have to be determined.

Considering the advantage of the second formula, the expression of Eq. (18) was adopted. The problem of a possible negative void ratio is due to the fact that the formula is based on a semilogarithmic plot. If a double-logarithmic plot is used, then Eq. (18) can be revised as

$$\log(e_c) = \log(e_{c0}) - \lambda\left(\frac{p'}{p_{at}}\right)^{\xi} \tag{20}$$

which can be rewritten as

$$e_c = e_{c0}\exp\left[-\lambda\left(\frac{p'}{p_{at}}\right)^{\xi}\right] \tag{21}$$

Given different values of the parameters e_{c0}, λ, and ξ, different curves can be obtained by Eq. (21), as shown in Fig. 7. It can be concluded that ξ controls the slope of the CSL after the inflection point [Fig. 7(a)], λ controls the position of the inflection point [Fig. 7(b)], and the combination of the three parameters gives access to an almost constant slope of the CSL as in Eq. (17) [Fig. 7(c)]. This enhanced formula gives the advantage of a nonlinear CSL, and at the same time imposes the positiveness of the void ratio with three parameters as input.

With the critical state concept, the density state of the soil can be defined by the ratio e_c/e or the distance $e_c - e$, where e is the current void ratio and e_c is the critical void ratio at the current stress

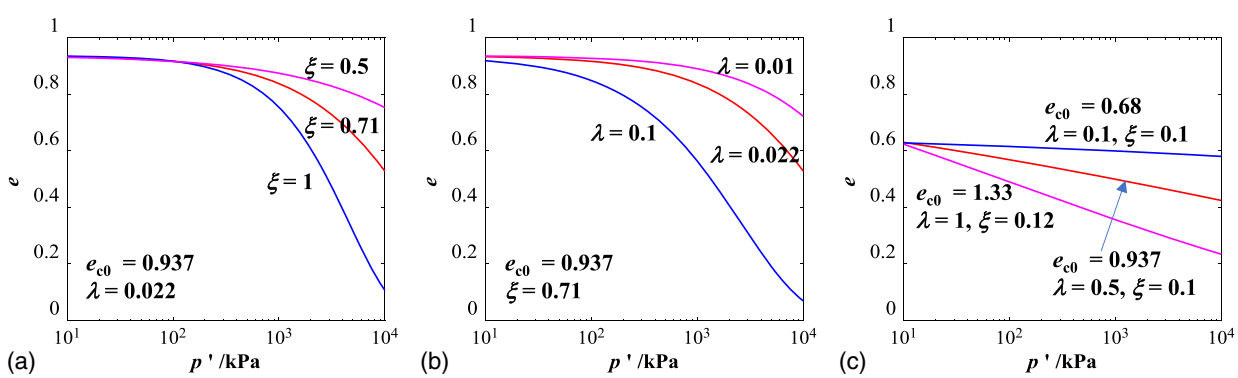

Fig. 7. Different curves of CSL described by the enhanced equation: (a) effect of ξ; (b) effect of λ; (c) effect of the combined parameters for an almost straight line

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state p', obtained from the equation of the CSL. This implies $e_c/e < 1$ or $e_c - e < 0$ for loose packing, $e_c/e > 1$ or $e_c - e > 0$ for dense packing, and $e_c/e = 1$ or $e_c - e = 0$ when the initial state lies on the CSL in the $e - p'$ plane. In agreement with experimental results obtained by Biarez and Hicher (1994), the density state effect to the stress-strain-strength behavior was implemented into the model by the following relation:

$$\phi_p = \arctan\left(\frac{e_c}{e}\tan\phi_c\right),$$
$$\phi_{pt} = \arctan\left(\frac{e}{e_c}\tan\phi_c\right) \tag{22}$$

The expression of the mobilized peak friction angle ϕ_p in Eq. (22) shows that, in a loose structure, ϕ_p is initially smaller than the critical friction angle ϕ_c. By contrast, a dense structure provides a higher degree of interlocking. Thus, ϕ_p is greater than ϕ_c. When the loading stress reaches the peak stress, a dense structure will dilate and the degree of interlocking will decrease. As a consequence, the peak friction angle will be reduced, which results in a strain-softening phenomenon.

The mobilized-phase-transformation angle ϕ_{pt} in Eq. (22) implies that in a loose structure with $e > e_c$, ϕ_{pt} is bigger than ϕ_c, which leads to a continuously contractive behavior; in a dense structure with $e < e_c$, ϕ_{pt} is smaller than ϕ_c, which imposes a dense structure to be initially contractive, and then dilative during a deviatoric loading.

For both loose and dense structures, when the stress state reaches the CSL, the void ratio e becomes equal to the critical void ratio e_c, and no volume change will take place any longer. Thus, the constitutive equations guarantee that stress and void ratio reach simultaneously the critical state in the $p' - q - e$ space. Eq. (22) gives a similar density effect as in the model suggested by Manzari and Dafalias (1997), but in a different form.

Using the aforementioned mobilized peak friction angle ϕ_p and the mobilized-phase-transformation angle ϕ_{pt}, and accounting for the Lode angle dependency, the peak strength ratio M_p and the phase-transformation stress ratio M_{pt} can be expressed by the following relations:

$$M_p = \left\{\begin{array}{ll} \dfrac{6\sin\phi_p}{3-\sin\phi_p} & \text{in compression} \\ \dfrac{6\sin\phi_p}{3+\sin\phi_p} & \text{in extension} \end{array}\right\},$$
$$M_{pt} = \left\{\begin{array}{ll} \dfrac{6\sin\phi_{pt}}{3-\sin\phi_{pt}} & \text{in compression} \\ \dfrac{6\sin\phi_{pt}}{3+\sin\phi_{pt}} & \text{in extension} \end{array}\right\} \tag{23}$$

To enhance the model, Eqs. (21)–(23) were introduced in the model presented in the previous section, and then used to simulate conventional drained and undrained triaxial tests on samples with different initial void ratios. As shown in Fig. 8, the enhanced model with the critical state concept can well reproduce the density effect on the mechanical behavior of granular materials.

Eq. (23) can also be applied to the basic model to enhance the strength in extension following the strength criterion of Coulomb or Matsuoka and Nakai (1974) without additional parameters.

Nonlinear Stress Dilatancy

Experimental investigations showed that the stress path in the $p' - q$ plane developed vertically at the beginning of an undrained triaxial loading, which implied that only a very slight volumetric strain was induced at the beginning of the shearing. The stress-dilatancy equation [Eq. (9)] should, therefore, be modified as follows:

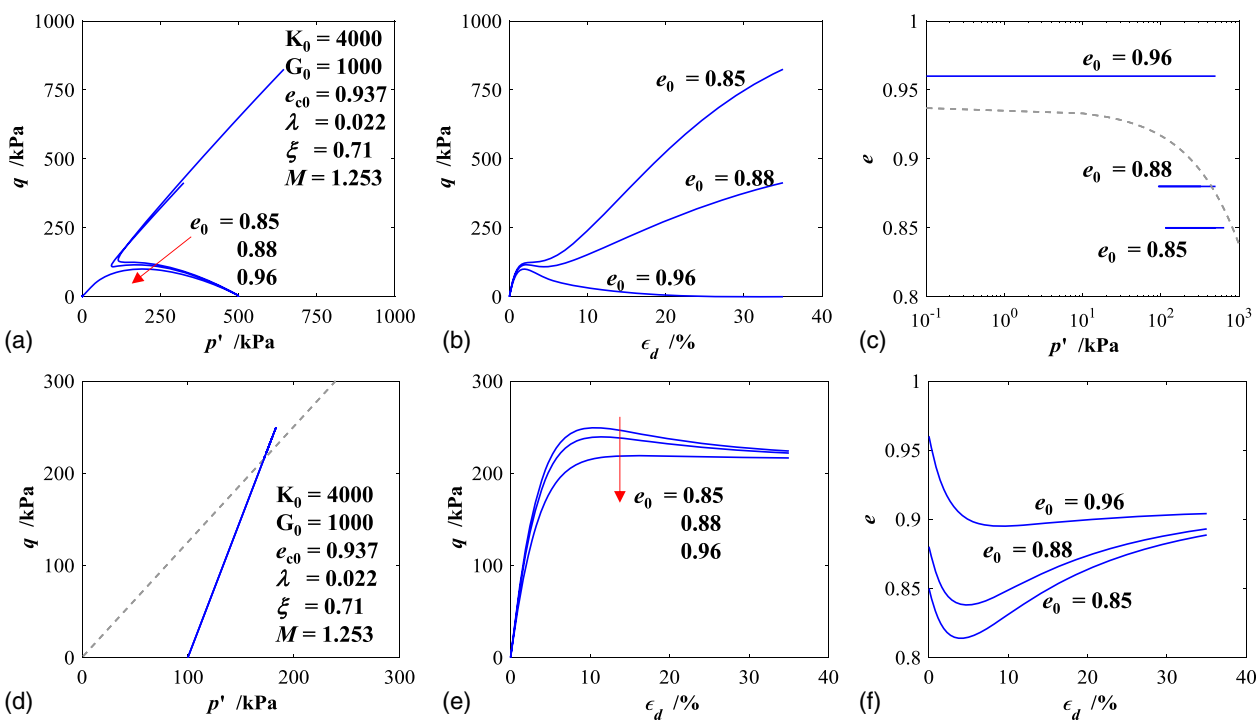

Fig. 8. Simulated results of (a–c) undrained triaxial tests with different initial void ratios; (d–f) drained triaxial tests with different initial void ratios

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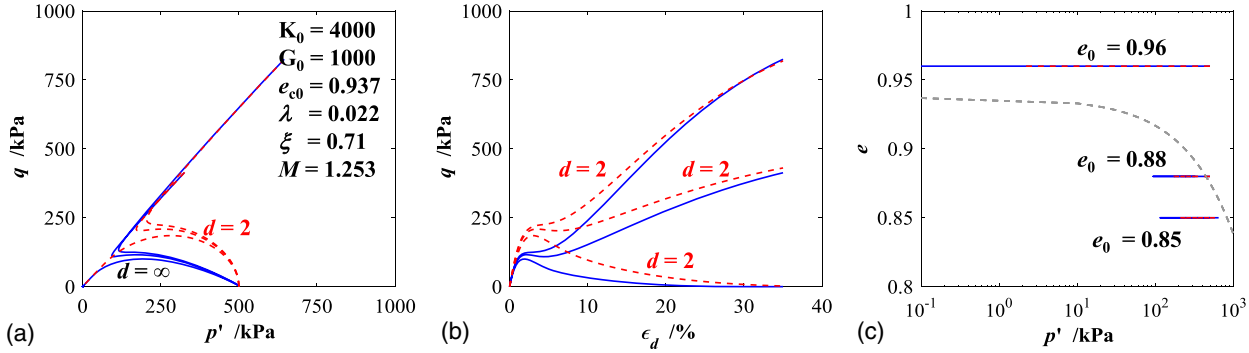

Fig. 9. Simulated results of undrained triaxial tests with different initial void ratios with the influence of a nonlinear stress dilatancy

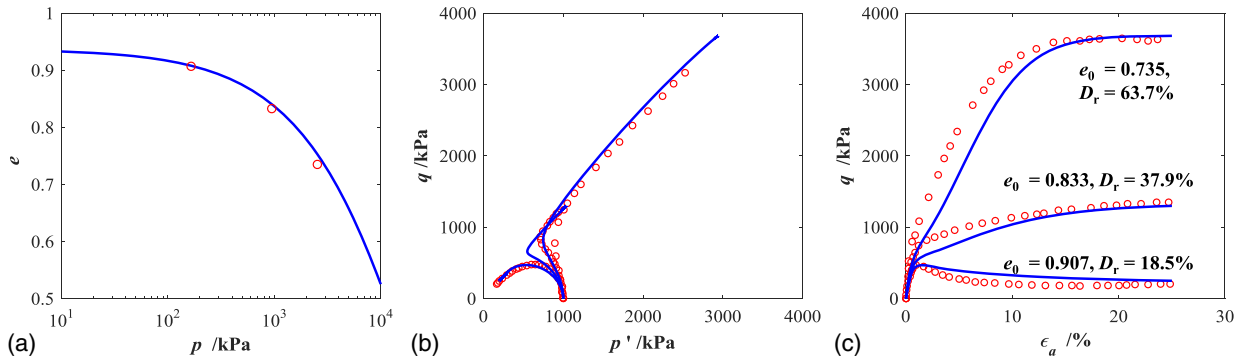

Fig. 10. Parameters determination from three undrained triaxial tests on Toyoura sand isotropically compressed at 1,000 kPa with different initial densities: (a) CSL in the $e - \log p'$ plane; (b) stress paths; (c) deviatoric stress versus deviatoric strain

$$\frac{\dot{\varepsilon}_v^{in}}{\dot{\varepsilon}_d} = \left(\frac{\dot{\varepsilon}_d}{|\dot{\varepsilon}_d|} M_{pt} - \frac{q}{p}\right)[1 - \exp(-d|\eta - \eta^R|)] \tag{24}$$

where η^R = stress ratio at the strain reversal state; and d = material constant controlling the nonlinearity of the stress dilatancy. The occurrence of a strain reversal is controlled by the condition $d\gamma^{(i-1)} \times d\gamma^{(i)} < 0$, with $\eta^R = \eta^{(i-1)}$ (i represents the current step, and $i-1$ the previous step). This nonlinear function was introduced into Eq. (11) as follows:

$$\begin{aligned} \dot{p}' &= K(\dot{\varepsilon}_v - \dot{\varepsilon}_v^{in}) \\ &= K\left\{\dot{\varepsilon}_v - \left(M_{pt}|\dot{\varepsilon}_d| - \frac{q}{p'}\dot{\varepsilon}_d\right)[1 - \exp(-d|\eta - \eta^R|)]\right\} \end{aligned} \tag{25}$$

The enhanced model was then used to simulate undrained triaxial tests on samples with different initial void ratios. As shown in Fig. 9, the introduction of the nonlinear stress dilatancy could improve the quality of the numerical simulations in terms of stress path and deviatoric stress-strain response during undrained loading.

Model Validation by Experimental Data

Different triaxial tests under both monotonic and cyclic loadings on Toyoura sand were selected for validating the model. Toyoura sand is a uniform fine quartz sand that consists of subrounded to subangular grains. It has been widely tested (Miura and Yamanouchi 1975; Miura et al. 1984; Pradhan 1990; Verdugo and Ishihara 1996; Yoshimine et al. 1998; Uchida and Stedman 2001). Toyoura sand has a maximum void ratio of 0.977, a minimum void ratio of 0.597, and a specific gravity of 2.65.

Three undrained triaxial tests performed by Verdugo and Ishihara (1996) under a confining stress of 1,000 kPa with different initial void ratios were used to calibrate the model parameters. The three parameters defining the CSL were measured from the position of the critical states in the $e - \log p'$ plane [Fig. 10(a)]. Common values of the elastic parameters $\upsilon = 0.3$ and $n = 0.67$ were adopted. The critical friction angle was measured from the stress paths in the $p' - q$ plane [Fig. 10(b)]. The shear stiffness parameter G_0 and the stress-dilatancy parameter d were obtained by curve fitting of three undrained tests [Figs. 10(b and c)]. The values of all the parameters are summarized in Table 1.

Triaxial Tests under Monotonic Loading

Verdugo and Ishihara (1996) conducted a series of monotonic drained and undrained triaxial tests on isotropically consolidated

Table 1. Values of Model Parameters for Toyoura Sand

Parameters	Values
G_0 (kPa)	3,000
υ	0.3
N	0.67
ϕ_c (degrees)	31.2
e_{c0}	0.937
λ	0.022
ξ	0.71
d	2

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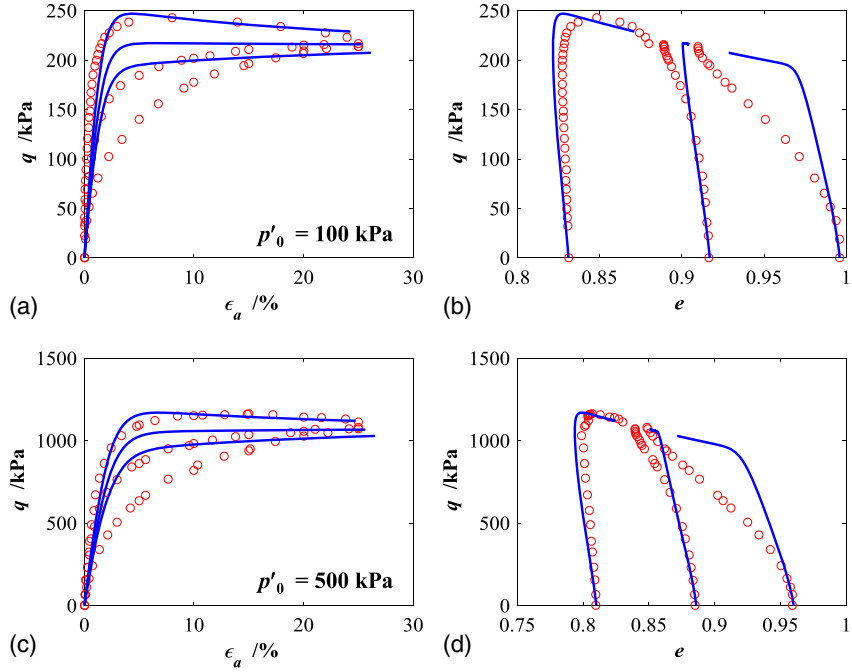

Fig. 11. Comparison between experimental and numerical results of drained triaxial compression tests on Toyoura sand with different initial void ratios under confining stresses of 100 and 500 kPa

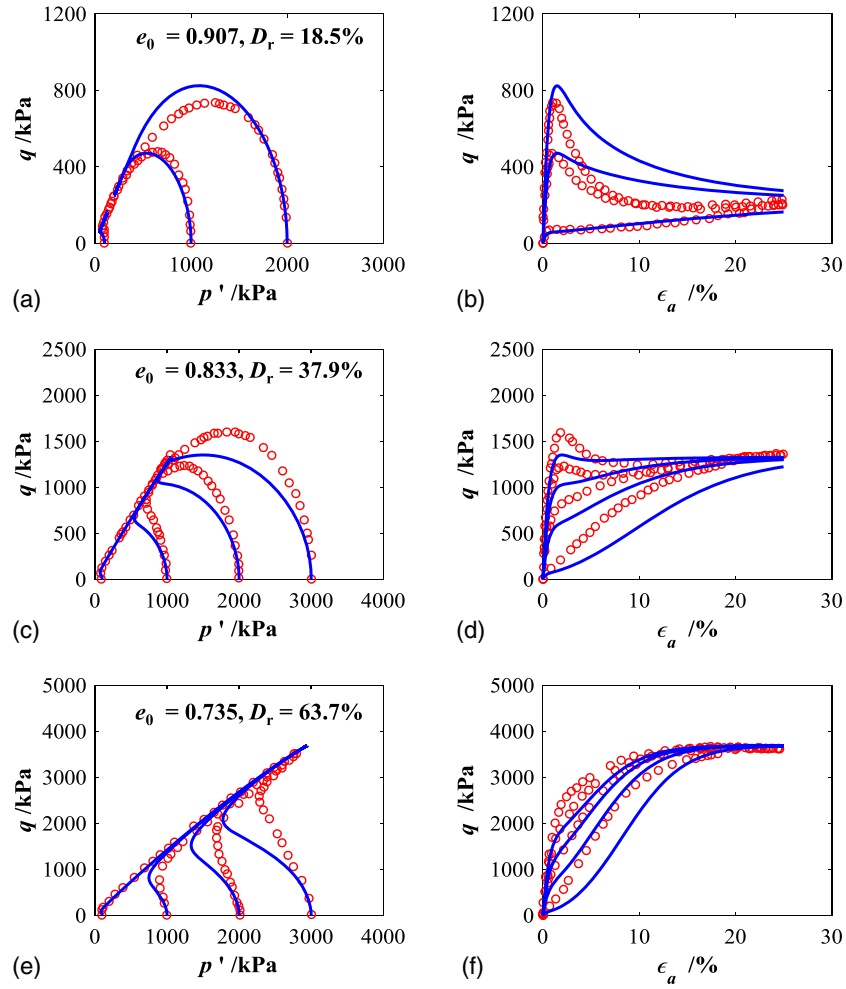

Fig. 12. Comparison between experimental and numerical results of undrained triaxial compression tests on Toyoura sand with different initial void ratios under confining stresses from 100 to 3,000 kPa

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samples of Toyoura sand with a mean diameter $D_{50} = 0.17$ mm and a uniformity coefficient $C_u = 1.7$: (1) drained triaxial tests with void ratios ranging from 0.810 to 0.996 under constant lateral stress of 100 and 500 kPa, and (2) undrained triaxial tests with void ratios ranging from 0.735 to 0.907 under different confining pressures from 0.1 to 3 MPa. These calibrated parameters were used to predict all these tests.

Fig. 11 shows the comparison between experimental data and numerical results for drained triaxial compression tests on isotropically consolidated samples. Fig. 12 shows the comparison between experimental data and numerical results for the undrained triaxial compression tests on isotropically consolidated samples of the same Toyoura sand. The tests covered an extensive range of confining pressures and void ratios. Very loose, loose, and medium sand were tested in conventional triaxial drained and undrained testing. The comparison between the test results and the model predictions shows an overall good agreement. For drained tests on very loose sand, the stiffness at small strains is overpredicted, which can be further improved by considering the feature of small strain stiffness.

Drained Cyclic Triaxial Tests

Pradhan (1990) conducted drained cyclic triaxial tests on Toyoura sand. However, the sand used by Pradhan (1990) came from a batch different from the one used by Verdugo and Ishihara (1996). Thus, the index properties were slightly different. The same values of the parameters determined from the tests of Verdugo and Ishihara (1996) were used to simulate the drained cyclic test, except for the value of $\phi_c = 32.3°$ in compression according to the drained test with loading–unloading and reloading.

Figs. 13(a–c) show the comparison between experiments and predictions for the drained cyclic triaxial test on loose Toyoura sand with an initial void ratio $e_0 = 0.845$ under constant $p' = 98$ kPa. Figs. 13(d–f) show the comparisons for a dense Toyoura sand with an initial void ratio $e_0 = 0.653$ under constant $p' = 98$ kPa.

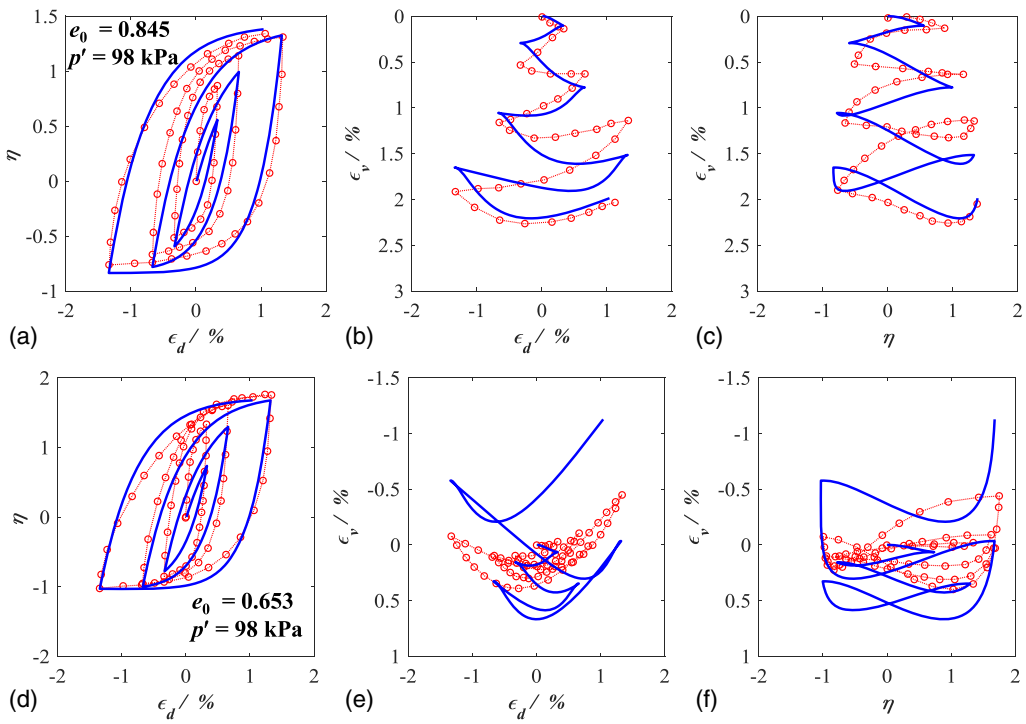

Fig. 13. Comparison between experimental and numerical results of constant-p' drained cyclic tests on Toyoura sand: (a–c) loose specimen; (d–f) dense specimen

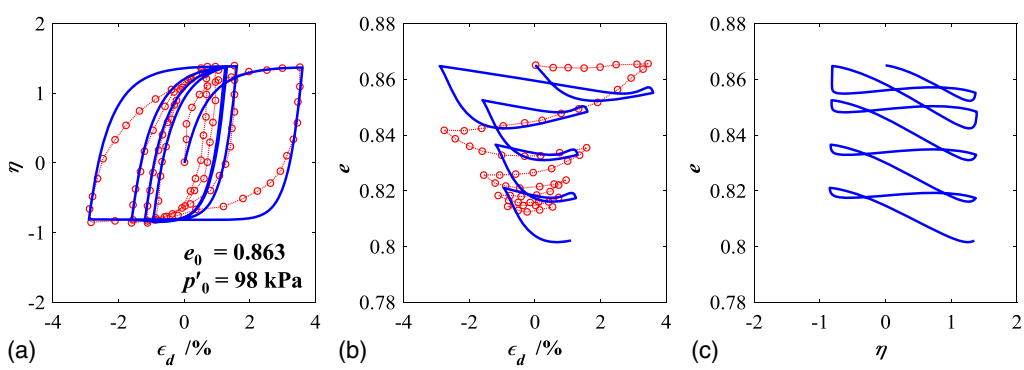

Fig. 14. Comparison between experimental and numerical results for drained cyclic triaxial tests on Toyoura sand

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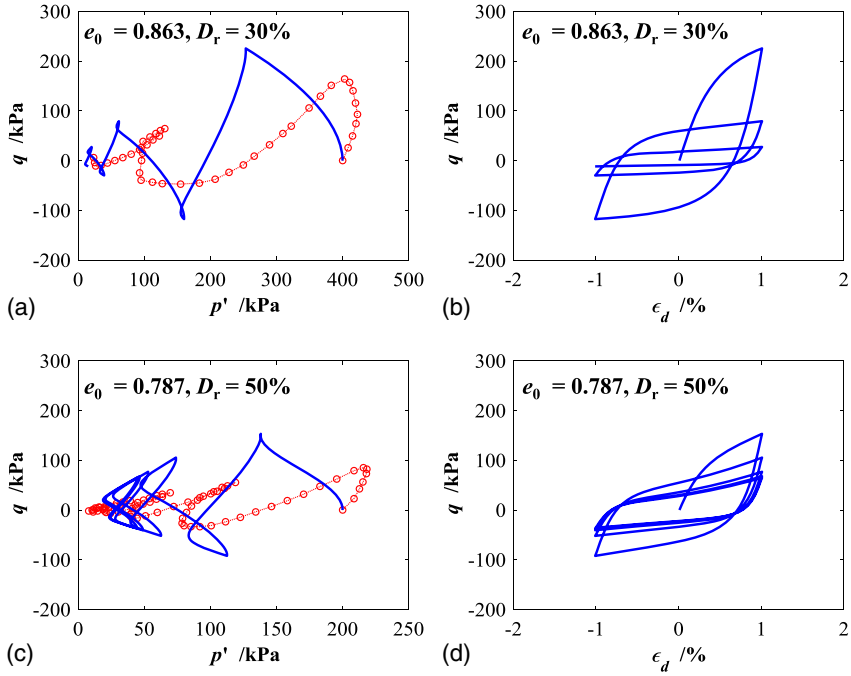

Fig. 15. Comparison between the experimental and numerical results of undrained triaxial tests under cyclic loading on Toyoura sand with different initial void ratios

Fig. 14 shows the comparisons between experiments and predictions for the drained cyclic triaxial test on loose Toyoura sand with an initial void ratio $e_0 = 0.863$ under constant confining stress $p_0' = 98$ kPa. A good agreement was achieved between experimental data and numerical simulations. The slight difference in the volumetric strain might be due to sample differences because the values of the parameters used for predicting this test were determined from a different test. The enhanced model captured the overall trend: at small strain amplitudes, loose sand densifies and dense sand dilates.

Undrained Cyclic Triaxial Tests

Uchida and Stedman (2001) conducted undrained cyclic triaxial tests on Toyoura sand. In each test, an axial strain of 1% magnitude was applied in each cycle until the liquefaction was obtained. The samples were prepared at two relative densities of 30 and 50% and subjected to two initial confining pressures of 200 and 400 kPa. The experimental and numerical results for the two samples are shown in Fig. 15. The model parameters calibrated from the monotonic tests were used for the predictions of the cyclic tests. The model reproduces the general trend as depicted by the test. Despite the difference in the evolution of the mean effective stress between experiments and simulations, which may be attributed to the slight difference in properties (like anisotropy) among sands from different batches and the lack of considering the internal fabric effect, the fact that a looser Toyoura sand had a greater liquefaction potential under higher confining pressure was well determined by the model.

Simulations of undrained cyclic tests at constant stress amplitude with $q_{\max} = 55$ kPa and $q_{\min} = -55$ kPa (noted as $q_{\text{cyc}} = 55$ kPa for symmetric loading condition) were conducted on samples with different initial void ratios using the aforementioned values of the model parameters. As expected, the dense sample exhibited a cyclic mobility phenomenon [Figs. 16(a and b)]; the medium dense sample also exhibited a cyclic mobility with higher deformations and reached an almost liquefied state in a fewer number of cycles than the dense sample [Figs. 16(c and d)]. The loose sample reached a complete liquefaction with large deformations in a few number of cycles [Figs. 16(e and f)]. Thus, the proposed model was able to predict the dependence of the liquefaction potential on the relative density of the sand.

Furthermore, for a given density ($e_0 = 0.865$) and the aforementioned set of parameters, simulations of undrained cyclic tests under different conditions were carried out with stress ratios ($q_{\text{cyc}}/p_0' = 0.12$, 0.24, 0.36) shown in Figs. 17(a and b), and with different types of cycles in terms of nonsymmetric loading condition (at cyclic deviatoric stress $q_{\text{cyc}} = 60$ kPa with different static deviatoric stresses $q_s = 0$, 30, and 80 kPa) shown in Figs. 17(c and d). Again, the model can capture the experimental trends (e.g., Yang and Sze 2011) for the effects of stress ratio and symmetric/nonsymmetric loading condition during undrained cyclic loadings.

For simulating more precisely undrained cyclic tests, the model still needs further enhancement, such as the intergranular strain effect suggested by Niemunis and Herle (1997).

Conclusion

A nonlinear incremental model based on a simple exponential function was first suggested. Three simulation schemes for modeling undrained triaxial tests, drained triaxial tests under constant p' and under constant confining stress, and cyclic tests under stress control were established. With only four parameters, the model formulated in axisymmetric condition demonstrated a good performance in reproducing basic features of a granular material under both monotonic and cyclic loading conditions.

Then, a nonlinear elasticity, a nonlinear stress dilatancy, and the critical state concept were introduced gradually to enhance the model. The capacity of the enhanced model with eight parameters was evaluated by simulating various triaxial tests on Toyoura sand under different conditions (i.e., undrained tests, drained constant-p' tests, drained constant confining stress tests, and monotonic and cyclic loadings). Overall, a good performance was achieved over all the simulations.

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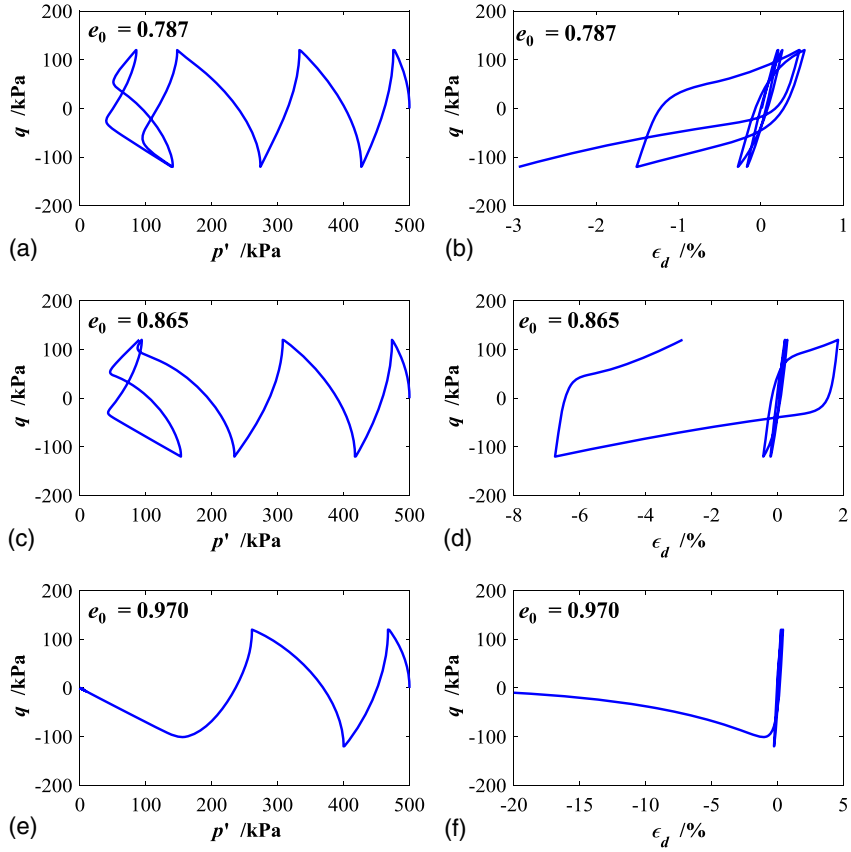

Fig. 16. Numerical results for undrained cyclic triaxial tests on Toyoura sand with three different initial void ratios

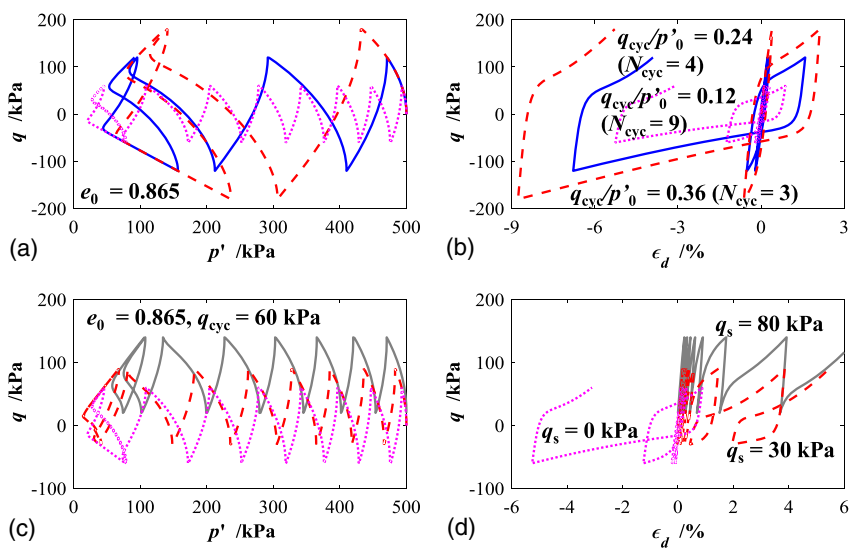

Fig. 17. Numerical results for undrained cyclic triaxial tests on Toyoura sand with (a and b) three different initial stress ratios; (c and d) three different static deviatoric stresses

Future work will focus on the extension of the model in the general stress space for solving boundary value problems, in which the physical consideration of fabric anisotropy (e.g., Li and Dafalias 2012; Gao et al. 2014; Dafalias and Taiebat 2014) should be incorporated.

Acknowledgments

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